

A. Bearings (Direction)

Definition: The angle measured clockwise from North.

Rules:

Always 3 digits (e.g., 045°, 008°, 180°).

North = 000°, East = 090°, South = 180°, West = 270°.

Back Bearing (Return Direction):

If Bearing < 180°, add 180°.

If Bearing > 180°, subtract 180°.

Example: Forward 050°, Back = 050° + 180° = 230°.

B. Navigation Problems (Ship/Plane Movement)

Simple Method (Using Right-Angled Triangle):

From a point, if you move on a bearing of θ :

North/South distance = Distance $\times$ cos θ

East/West distance = Distance $\times$ sin θ

Tip: Draw a North arrow first. Then draw the bearing angle. Form a right-angled triangle.

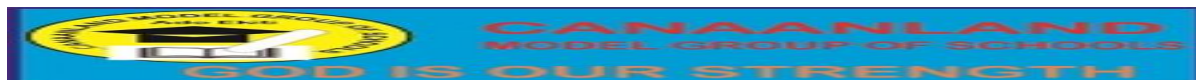
C. Chord and Arc Properties (Circle)

Radius (r): Distance from center to edge.

Chord (c): Straight line joining two points on a circle.

Central angle (α): Angle at the center.

Key Formulas (Memorize):



Chord length: $c = 2r \times \sin(\alpha/2)$

Arc length: $s = (\alpha/360^\circ) \times 2\pi r$

Perpendicular distance (h) from center to chord: $h = r \times \cos(\alpha/2)$

D. Complex Scenarios (Two or More Legs)

Strategy for Multi-stage Journey:

Draw diagram for each stage.

Find total East-West distance (add/subtract carefully).

Find total North-South distance.

Use Pythagoras to find direct distance.

Use $\tan \theta = (\text{total East or West}) / (\text{total North or South})$ to find bearing.

Worked Examples (Short & Clear)

Example 1 (Bearings):

A ship sails 40 km on bearing 065° . How far East and North?

East = $40 \times \sin 65^\circ = 40 \times 0.9063 = 36.25$ km

North = $40 \times \cos 65^\circ = 40 \times 0.4226 = 16.90$ km

Example 2 (Chord Length):

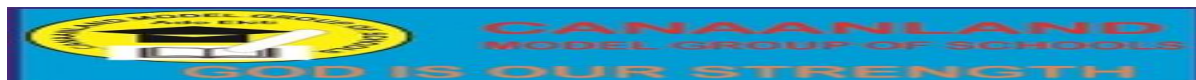
Radius = 8 cm, central angle = 120° . Find chord length.

Chord = $2 \times 8 \times \sin(120^\circ/2) = 16 \times \sin 60^\circ = 16 \times 0.8660 = 13.86$ cm

Example 3 (Distance between two ships):

Ship A: 30 km at 040° , Ship B: 25 km at 130° from same port.

A: East = 19.28 km, North = 22.98 km



B: East = 19.15 km, South = 16.07 km

Difference: East = 0.13 km, North-South = 39.05 km

Distance = $\sqrt{(0.13^2 + 39.05^2)} = 39.05$ km

Example 4 (Arc Length):

Radius = 12 cm, angle = 75° . Find arc length.

Arc = $(75/360) \times 2 \times \pi \times 12 = (75/360) \times 24\pi = 5\pi = 15.71$ cm

Class Work

Using a protractor, plot a journey: A $\rightarrow$ B on bearing 045° for 5 cm (scale 1 cm = 10 km). Then B $\rightarrow$ C on bearing 135° for 4 cm. Find direct distance from A to C.

Calculate chord length for radius 10 cm, central angle 90° .

Two students start from same point. One walks 4 m East, the other 3 m North. Find distance between them.

Questions (5 minutes)

A boat travels 50 km on bearing 075° . Find how far North and East.

Chord length = 16 cm, radius = 10 cm. Find the central angle.

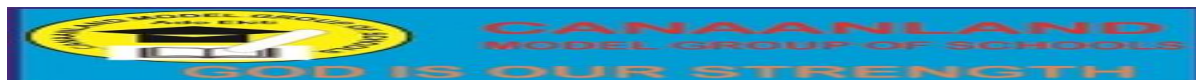
A ship sails 30 km at 120° , then 40 km at 210° . Find its direct distance from start.

Radius = 15 cm, arc length = 18 cm. Find the central angle in degrees.

Two towns A and B are 100 km apart. Town C is 60 km from A on bearing 040° . Find the bearing of B from A. (Use diagram)

Assignment (Take Home)

Plane flies A $\rightarrow$ B: 300 km at 065° , then B $\rightarrow$ C: 200 km at 155° . Find:



(a) Distance A to C

(b) Bearing of C from A

Circle radius 20 cm, chord angle = 100° . Find:

(a) Chord length

(b) Perpendicular distance from center to chord

5.

Two ships leave harbor: P at 25 km/h bearing 040° , Q at 20 km/h bearing 130° . After 2 hours, find:

6.

(a) Distance between them

(b) Bearing of Q from P

A person walks 8 km at 200° , then 6 km at 290° . Find:

(a) How far West from start

(b) Direct distance from start

Radius = 14 cm, central angle = 150° . Find:

10.

(a) Arc length

(b) Sector area

(c) Chord length

Summary for Board/Display (Key Points)

Concept Formula / Rule

Bearing Clockwise from North, 3 digits

Back Bearing $\theta \pm 180^\circ$

East/West distance Distance $\times \sin \theta$

North/South distance Distance $\times \cos \theta$

Chord length $2r \times \sin(\alpha/2)$





Arc length $(\alpha/360) \times 2\pi r$

Final distance $\sqrt{[(\text{total E/W})^2 + (\text{total N/S})^2]}$

Golden Rule for Navigation: Always draw the North arrow first. Draw the bearing angle from North. Then solve the triangle.

